

Lab Exercise: Python Slicing With Steps

Objective

Practice slicing a Python tuple to extract subsets of items, including slices with **start**, **end**, and **step** parameters.

Given Data

```
items = ("shoes", "gloves", "hats", "shirts", "blouses", "t-shirts", "pants")
```

Part 1 — Basic Slices

Use the tuple above for each task.

- **First_two_items** — Slice indices 0 through 2

```
items[0:2]
```

Expected: ("shoes", "gloves")

- **Middle_three** — Extract "hats", "shirts", "blouses"

```
items[2:5]
```

Expected: ("hats", "shirts", "blouses")

- **Last_two** — Use negative indices

```
items[-2:]
```

Expected: ("t-shirts", "pants")

Part 2 — Slicing With Step

These tasks focus on the full slice form:

```
items[start:end:step]
```

- **Every_second_item** — Use step = 2

```
items[0:7:2]
```

Expected: ("shoes", "hats", "blouses", "pants")

- **Step_from_middle** — Start at index 1, step = 2

```
items[1:7:2]
```

Expected: ("gloves", "shirts", "t-shirts")

- **Reverse_every_second** — Use a negative step

```
items[::-2]
```

Expected: ("pants", "blouses", "shirts", "gloves")

Part 3 — Mini-Challenge

Write one slice that returns:

- "gloves", "shirts", "t-shirts"
- Using **only one slice**
- Using **step = 2**

(Hint: start at index 1)

The challenge asked you to produce a slice that returns:

- "gloves"
- "shirts"
- "t-shirts"

Using **one slice** and **step = 2**.

✓ Solution

```
items[1:7:2]
```

Why it works

- Index **1** → "gloves"
- Step **2** → skip every other item
- Slice continues through index **6** → "t-shirts"

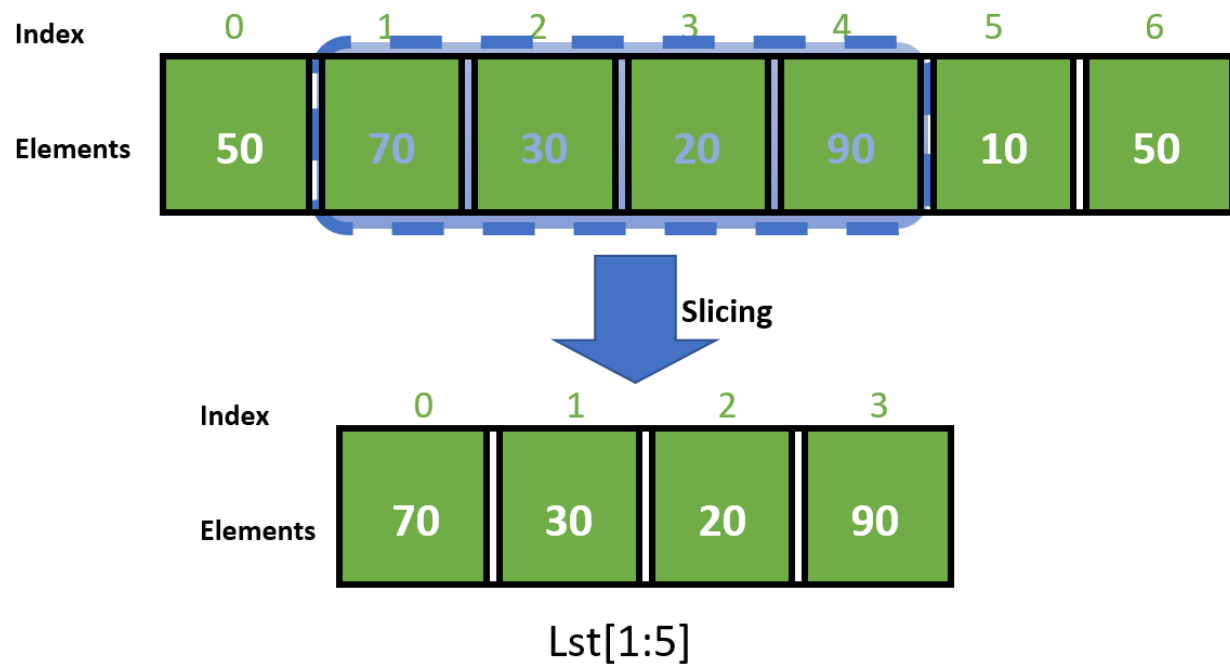
So the slice walks the tuple like this:

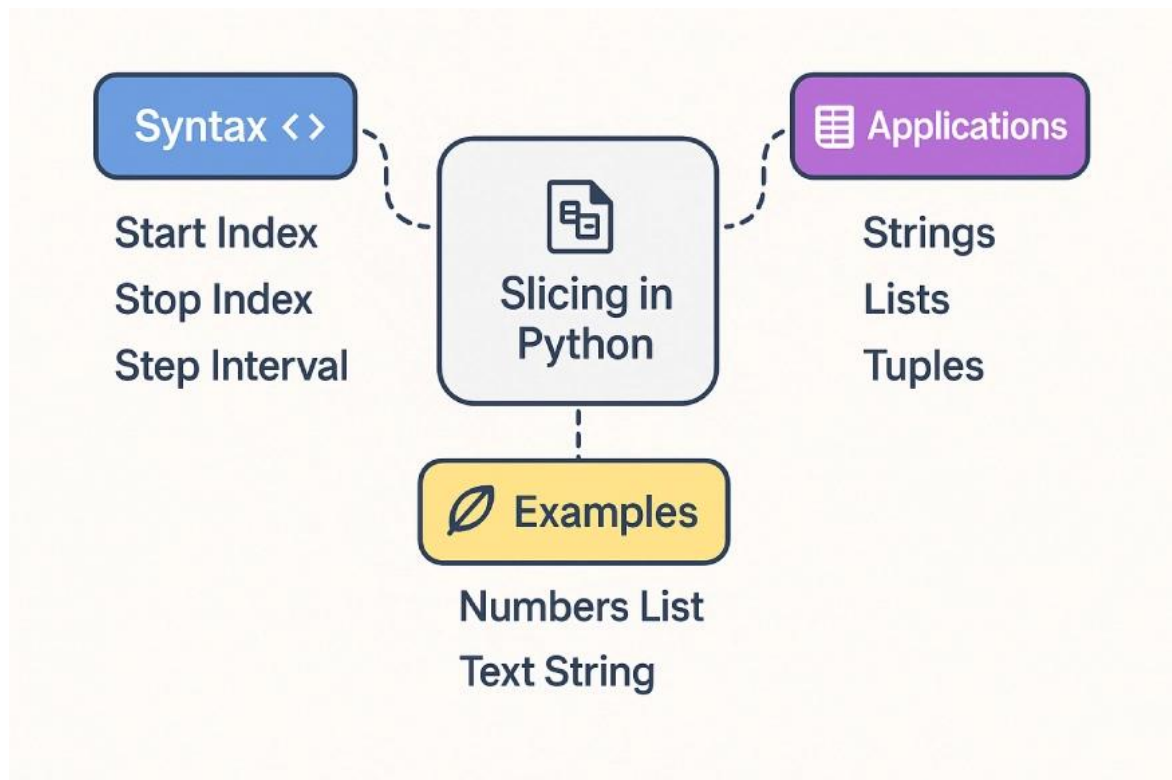
Code

```
1 → "gloves"
3 → "shirts"
5 → "t-shirts"
```

Result

```
python  
("gloves", "shirts", "t-shirts")
```





Slicing in Python

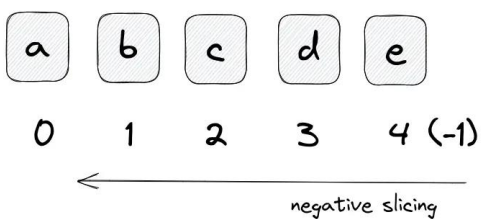
Definition

extracting a specific portion of a sequence

Syntax

[start:end:step]

[1:4:-1]



Examples

list

```
letters = ['a', 'b', 'c', 'd', 'e']
letters[1:3]
```

string

```
text = "Indexing and slicing"
print(text[4:10])
```

reverse list

```
letters = ['a', 'b', 'c', 'd', 'e']
letters[::-1]
```

[e, d, c, b, a]

Slicing works on

- * string
- * list
- * tuple
- * array

- * pandas
- * numpy

Indexing vs slicing

letters[2]

vs

letters[1:3]